

3.3) SUBSET SELECTION

- Prediction accuracy
- Interpretation

Objective: Select a subset of $\hat{\beta} = (\beta_0, \beta_1, \dots, \beta_p)$

3.3.1) BEST-SUBSET SELECTION

- Find the subset of size k ($k \in \{0, 1, \dots, p\}$) that gives the smallest residual sum of squares:

$$RSS = \sum_{i=1}^n (y_i - X\beta)^2 \quad \begin{array}{l} \dim(X) = n \times k \\ \dim(\beta) = k \end{array}$$

- Computationally expensive. Why?

How many ways are there of selecting k elements from p elements?

$$\binom{p}{k} = \frac{p!}{k!(p-k)!} \quad n! = n \cdot (n-1) \cdot \dots \cdot 2 \cdot 1$$

→ 'p choose k'

3.3.2) FORWARD- AND BACKWARD-STEPWISE SELECTION

- Sequentially add into the model the predictor that most improves the fit. (Forward)
- Sequentially subtract from the model the predictor that most preserves the fit. (Backward). Works by dropping the lowest z-score.

* Computationally more efficient, greedy algor: lhm, usually similar performance to best subset.

3.4) SHRINKAGE METHODS

Continuous version of subset selection.

3.4.1) RIDGE REGRESSION

- Shrinks coefficients by imposing a penalty on their size.

$$\hat{\beta}_R = \underset{\beta}{\operatorname{argmin}} \left(\sum_{i=1}^N (y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right)$$

$$\beta = (\beta_1, \beta_2, \dots, \beta_p) - \text{Exclude } \beta_0$$

λ : Shrinkage parameter

Equivalently,

$$\hat{\beta}_R = \underset{\beta}{\operatorname{argmin}} \left(\sum_{i=1}^N (y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j)^2 \right), \quad \sum_{j=1}^p \beta_j^2 \leq t$$

*) There is a 1-1 correspondence between t and λ .

We normalise the inputs. Can you tell why? How do we exclude β_0 ? Two ways:

1) Remove the column of 1s from X and do:

$$x_j = \frac{X_{ij} - \mu_{x_j}}{\sigma_{x_j}}, \quad y = y - \mu_y$$

$$\boxed{\begin{array}{l} \mu_{x_j} = \frac{1}{N} \sum_{i=1}^N x_{ij}, \quad \sigma_{x_j}^2 = \frac{1}{N} \sum_{i=1}^N (x_{ij} - \mu_{x_j})^2 \\ \mu_y = \frac{1}{N} \sum_{i=1}^N y_i \end{array}}$$

$$\text{Then } \hat{\beta} = (X^T X + \lambda I)^{-1} X^T y \quad (I \text{ has dim } p \times p)$$

$$\hat{\beta}_0 = \mu_y - \mu_x^T \hat{\beta} \quad (\mu_x \text{ has dim } p)$$

2) Change I for $D = \begin{bmatrix} 0 & 0 & \dots \\ 0 & 1 & 0 \\ \vdots & 0 & 1 \end{bmatrix}$, then

$$\hat{\beta} = (X^T X + \lambda D)^{-1} X^T y \rightarrow \beta_0 \text{ included}$$

3.4.2) THE LASSO

$$\hat{\beta}_L = \underset{\beta}{\operatorname{argmin}} \left[\sum_{i=1}^N (y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j)^2 + \lambda \sum_{j=1}^p |\beta_j| \right]$$

- No analytical solution, gradient descent?

Exercises (Certificate of engagement)

1) Improve the LinearRegression class from Session 1 to include:

- Best subset selection
- Forward & BACKWARD-Stepwise selection
- Ridge regression

2) Try your class methods on the three datasets:

- What results do you obtain
- Which one works best? Why?

3) Implement the solution for the Lasso using gradient descent (Hard).